

1. Company Situation Analysis

Company Overview

The North-West University Protection Services is responsible for keeping all three university campuses safe and secure. Their duties are quite broad and include protecting university property, controlling access and traffic, investigating incidents and making sure the environment is safe for students, university staff and visitors.

As part of these security operations, the Protection Services also manages a lost and found service. This service is based at their specific buildings on each campus, Building A16 at the Mahikeng Campus, Building F17 at the Potchefstroom Campus and Building F11 at the Vanderbijlpark Campus.

At the moment, the lost and found system is completely manual. When a student or university staff finds an item that they believe to be lost on campus, they physically hand it over to Protection Services staff. The Protection Services staff then record the item details in a physical logbook and store the item safely. If students or university staff lose something, they have to go to the Protection services building in person to check whether their item has been found.

Objectives

The main purpose of the current physical lost and found service is to make sure that all items found on campus are safely collected, properly recorded in logbooks, securely stored and eventually returned to the correct owners.

The system is also designed to ensure full accountability for all items that are kept by Protection Services. Items are only given back once the owner has been properly verified in person, which helps prevent theft or false claims.

Operations

The current system works separately at each campus and follows a fully physical process. When a student or university staff member finds an item anywhere on campus, for example in a bathroom, lecture room or outside on campus grounds, they take it to the Protection Services building for that specific campus.

The Protection Services staff then receive the item and manually record its details in a logbook. This usually includes a short description of the item and the date it was handed in. After that, the item is stored in a secure storage room.

If someone is looking for a lost item, they must go in person to the Protection Services building and ask if their item has been handed to the Protection Services staff. The Protection Services staff then manually check through the written logbooks and physically go through the stored items to see if there is a possible match.

If a match is found, the person claiming the item must provide enough proof of ownership in person. Once the item has been verified, it is handed back to the owner and the logbook is updated to show that the item has been successfully claimed.

Organizational Structure

The lost and found service functions entirely within the structure of the Protection Services. The Protection Services staff serve as the main authority and are the only people responsible for managing the process. Their duties include receiving found items, recording details, storing the items securely and verifying ownership claims, all while also handling their campus security responsibilities.

Within this structure there is a hierarchy. Campus specific Protection Services staff are responsible for managing day to day operations at their respective campus buildings, including the recording, updating and handling of lost and found items. Senior Protection Services management sits above this operational level and is responsible for overseeing activities across all three university campuses.

Students and university staff are external users of the system. They do not take part in managing or updating any records. Their involvement is limited to physically handing in items they have found or visiting the Protection Services building to check and claim lost belongings.

The Protection Services buildings are Building A16 at Mahikeng Campus, Building F17 at Potchefstroom Campus and Building F11 at Vanderbijlpark Campus, they are the central points where all lost items are received, stored and processed for claims.

2. Problems and Constraints

Problems

The current manual system has several serious operational and administrative challenges.

The main issue is that there is no central digital database, which leads to fragmented information. As a result, it is not possible to search for lost items on each campus.

Students and university staff also need to physically go to the Protection Services buildings, sometimes more than once, just to check whether their lost item has been found. This makes the process inefficient and time consuming.

In addition, the Protection Services staff already have many other security responsibilities. Because of this, having to manually search through physical logbooks to locate lost items adds a significant and unsustainable administrative workload.

Also the current system has a complete lack of transparency and visibility for the users of the

service, that is students and university staff. Many items remain unclaimed and end up accumulating in storage because owners often assume their belongings were never recovered. In other cases, first year students are not aware of the lost and found service and how the lost and found service works, so they do not follow up and eventually give up looking for their items.

Another major limitation is that the use of physical logbooks keeps all information isolated within each campus building. This makes it impossible to combine or analyse data across campuses. As a result the university cannot perform any meaningful data analysis to identify patterns.

Constraints

The proposed system is limited to use only within the university campuses and is intended exclusively for verified students and university staff, which reflects how the current lost and found service works.

Importantly, it will not replace the existing physical lost and found processes. All physical items will still need to be handled, stored and claimed in person at the relevant Protection Services buildings, which are Building A16 at the Mahikeng Campus, Building F17 at the Potchefstroom Campus and Building F11 at the Vanderbijlpark Campus.

As with the current lost and found process, only authorized Protection Services staff will be allowed to capture item records, approve claims and update the status of items in the system. The system must also comply with strict data privacy requirements, meaning that sensitive personal information such as student numbers and claimant contact details must remain hidden from public access.

In addition, the system will only display the information that is provided by the Protection Services staff. This also places a restriction on the Protection Services staff, ensuring that they do not upload overly specific identifying details or images that could interfere with or weaken the required physical verification during the claiming process.

3. Database System Specifications

Objectives of the proposed system

The proposed system aims to introduce a centralized digital inventory that acts as a support to improve the efficiency of the Protection Services lost and found service, without replacing their existing physical processes.

Its main goal is to allow the Protection Services staff to digitally record found items from all three campuses into a single, unified database that can be searched easily. This replaces the current fragmented, physical based logbook system with a more organized digital approach.

For students and university staff, the system will provide an online access platform where they can search and filter through listed found items to check whether their belongings have been recovered by Protection Services. This reduces the need for unnecessary physical visits to the campus security offices and improves overall accessibility to the service.

Furthermore, implementing a single consolidated platform across all campuses enables more data analysis, which cannot be done with the current way of doing things. It allows management to generate both cross campus and campus specific reports, which makes it possible to identify patterns such as frequently lost item types and areas where items tend to get lost and so many other aspects of analytics that were previously not visible due to the separation of physical logbooks.

Overall, the system is intended to significantly reduce the administrative burden on Protection Services staff by minimizing manual record searching and logbook handling. It should also help reduce the accumulation of unclaimed items by improving visibility of available lost property to students and university staff. It also strengthens accountability by tracking which authorized Protection Services staff member records, processes or releases each item.

Information required from the database

To function effectively, the database must store structured information for each found item. This includes the item name, a safe or non-sensitive description as provided by the Protection Services staff, the item category, the date it was received and the specific campus etcetera. The system may also support limited detail images as provided by the Protection Services staff, of items where appropriate.

The database must also capture operational metadata, such as which Protection Services staff member logged the item and which staff member verified and approved the claim. This is necessary for accountability and audit purposes.

The system should retain private claim information for internal use only. This includes the claimant's name, student or university staff number and the date the item was collected. This information must remain restricted and not publicly accessible.

Finally, the system must manage the full lifecycle of each item by tracking its status, from being available in storage to being successfully claimed and released to the rightful owner.

Scope of the system

The scope of the system includes the digital recording, storage, tracking and visibility of found items across the Mahikeng, Potchefstroom and Vanderbijlpark campuses.

It provides a digital interface for Protection Services staff to manage the inventory of items currently in their possession. To reflect the organizational structure, the system will use role based access control. The campus specific Protection Services Staff will be limited to recording, managing and updating item statuses only within their assigned campus. Their access is restricted to ensure they can only interact with data relevant to their local operations. An all campuses administrator role is intended for senior Protection Services management, who will have full access across all campuses. This includes the ability to view all the items and generate all campuses reports and analytics.

The system also provides a read only search interface for students and university staff, which will allow them to view and search for lost items across the university, they will be able to filter items by category, campus e.t.c. It also covers the complete lifecycle of each item, starting from the point it is digitally recorded by the Protection Services staff and ending when its status is updated to “claimed” after successful physical verification and collection.

Boundaries of the system

The boundaries of the system are defined by the operational procedures of the lost and found service as done by the Protection Services. The system is fully controlled by the Protection Services staff, while students and university staff members are limited to viewing and searching functionality only. They cannot submit lost item reports or modify any data, which ensures that the system remains a verified inventory of lost items rather than an unregulated reporting platform.

The system does not replace the physical claiming process in any way. It functions only as a visibility layer, meaning that no online or automated claiming is allowed. All ownership verification and item collection must still take place in person at the relevant Protection Services building.

The important part of the system is that it only displays the information that is entered by Protection Services staff. The Protection Services staff is responsible for ensuring that item descriptions and images are sufficiently generic to assist students and university staff identify that their item has been recovered, but not so detailed that they could enable fraudulent claims or expose sensitive data, for example when a student card is lost, they could blur the full student number and only show the first four number and the picture. Ultimately it is up to the Protection Services staff to identify what information is sensitive or unique about the lost item, there is no way of providing a way on how information must be entered since items differ, the system's job is to display and record information.